

Darren Zheng

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🌐 Personal Website: darrenzheng24.github.io

EDUCATION

University of Pennsylvania, College of Arts and Sciences

Bachelor of Arts in Mathematics, Minor in Statistics, Ben Franklin Scholar

August 2021 - May 2025

B.A. GPA: 3.51/4.0

PROJECTS

Independent Study

September 2025 - Dec 2025

- Reading "Fourier Analysis & Hausdorff Dimension" by Pertti Mattila under Prof. Phillip Gressman
- Short write-up regarding the proof of the Wolff-Erdogan Theorem can be found on my personal website [here](#)

Undergraduate Honor's Thesis | [Link to PDF](#)

June 2024 - May 2025

- Studied intrinsic Morse functions on Grassmannian manifolds under Prof. Dennis DeTurck

Directed Reading Program | [Link to Project Write-ups](#)

Fall 2022 - May 2025

- Semester-long independent study with a graduate student. Prepared a presentation/write-up at the end of each semester to rest of participants and mentors. Participated over 6 semesters
 - * Knot Invariants using Algebraic Topology (Yi Wang)
 - * Understanding Generating Functions (Xinxuan Zhang)
 - * Pontryagin Duality of LCA Groups (Travis Leadbetter)
 - * The Distinct Distances Problem (Jae Ho Choi)
 - * Besicovitch Sets & Hausdorff Dimension (Hunter Stufflebeam)
 - * Studying Mapping Class Groups (Ellis Buckminster)

EXPERIENCES

Penn Math Contest Club | *Co-founder & President*

Fall 2021 – May 2025

- Co-founded club during freshmen year, focused on hosting a math contest for high schoolers
- Wrote original problems for the test, organized day-of logistics and events, conduct outreach with local Philadelphia schools, speak to sponsors to secure funding, and grow the club for long-term sustainability
- Spring 2025 contest had ~150 in-person and ~200 online participants; [More details on website here](#)

Penn Undergraduate Math Society | *President*

Spring 2021 – May 2025

- Joined board as freshmen liaison, president during senior year. Help run events for all math undergraduate majors and broader community
- Organized events include: Course & Major Advising, Graduate Student talks, Integration Bee, Department Merch Sale, Origami Night, Weekly Cookie Office Hours, DRP x PUMS mini-talks, faculty lunches, End-of-Semester Socials

Putnam Club | *Organizer*

Fall 2021 – Fall 2024

- Organized club that meets weekly on Sundays in the fall to practice for the Putnam Exam

WORK

Ross Mathematics Program | *Admin*

Summer 2024

- Was an administrator during the 6-week math camp for high schoolers
- Handled various organizational tasks, coordinating social events for the students, and preparing various mathematical talks, including an week-long introduction to point set topology talk series for the students

AwesomeMath Summer Program | *Teaching Assistant*

Summer 2022/2023/2025

- Was a TA for three summers. Did 2 courses per each 3-week session for 9 weeks each summer
- Hosted problem sessions, office hours and graded daily HW for classes between 10-30 students
- Taught primarily combinatorics I/II, algebra III/IV, and number theory III/IV. See specific course details [here](#)

TEACHING

UPenn Math Teaching Assistant | *Teaching Assistant*

Fall 2022 – May 2025

- Responsible for grading assignments/exams, weekly office hours and recitations for most classes
 - * Grader for MATH3200 - Linear Algebra (Spring 2025)
 - * TA for MATH1410 - Calculus II (Fall 2024)
 - * TA for MATH2600 - Honors Calculus II (Spring 2024)
 - * TA for MATH1610 - Honors Calculus I (Fall 2023)
 - * TA for MATH1410 - Calculus II (Fall 2022)

VOLUNTEERING

Science Olympiad at Penn | *Test Writer*

Fall 2021 – May 2025

- Write tests for It's About Time, Fermi Problems, Optics, and Codebusters categories for high school competitors over 4 years
- Joined finance board during 2023-24 to secure sponsorship funding for the high-school contest

Matriculate | *Mentor*

Fall 2022 – Spring 2024

- Mentor first-generation/low-income high school seniors through the year-long college admissions process
- Scheduled weekly meetings and have overseen 3 cohorts (12 students in total)

Princeton Prison Teaching Initiative | *Grader*

Spring 2024

- Graded HW + Exams for a statistics class

Bridge to Math | *Volunteer*

Fall 2022 - May 2025

- Hosted some additional quiz/finals prep sessions throughout the semester as organized by Bridge-to-Math club @ Penn

AWARDS

Good Teaching Award for MATH2600

Spring 2024

Good Teaching Award for MATH1610

Fall 2023